**CHENNAI INSTITUTE OF TECHNOLOGY, CHENNAI - 600069**

*(An Autonomous Institution Affiliated to Anna University, Chennai)*

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

# CLEARDRIVE AI: REAL-TIME ADAS AND VISION RESTORATION FOR ADVERSE DRIVING CONDITIONS

**A PROJECT BASED LEARNING REPORT**

*Submitted by*

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## BONAFIDE CERTIFICATE

This is to certify that the Project-Based Learning report titled **"CLEARDRIVE AI: REAL-TIME ADAS AND VISION RESTORATION FOR ADVERSE DRIVING CONDITIONS"** is a bonafide record of work carried out by **SHIVANESH V (Reg. No: 210423243050)** and **CO-RESEARCHER (Reg. No: 210423243051)** of the Department of Artificial Intelligence and Machine Learning, Chennai Institute of Technology, as part of the continuous, mentor-guided Project-Based Learning (PBL) component of the **Machine Learning** course during the academic year **2026-2027** under my supervision.

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# LIST OF ABBREVIATIONS

| | | | |
|---|---|---|---|
| **ADAS** | Advanced Driver Assistance Systems | **HUD** | Heads-Up Display |
| **AIS** | Automotive Industry Standards (India) | **INT8** | 8-bit Integer Quantization |
| **ASIL** | Automotive Safety Integrity Level (ISO 26262) | **IPM** | Inverse Perspective Mapping |
| **BOM** | Bill of Materials | **JSON** | JavaScript Object Notation |
| **CLAHE** | Contrast Limited Adaptive Histogram Eq. | **LiDAR** | Light Detection and Ranging |
| **CMOS** | Complementary Metal-Oxide Semiconductor | **LWIR** | Long-Wave Infrared (8-14 um) |
| **CNN** | Convolutional Neural Network | **MAE** | Mean Absolute Error |
| **C-V2X** | Cellular Vehicle-to-Everything (3GPP Rel-16) | **mAP** | Mean Average Precision |
| **DCP** | Dark Channel Prior | **MJPEG** | Motion Joint Photographic Experts Group |
| **DMS** | Driver Monitoring System | **MoRTH** | Ministry of Road Transport & Highways |
| **DSRC** | Dedicated Short-Range Communications | **NMS** | Non-Maximum Suppression |
| **EAR** | Eye Aspect Ratio | **ODD** | Operational Design Domain |
| **EMA** | Exponential Moving Average | **PBL** | Project-Based Learning |
| **FLIR** | Forward-Looking Infrared | **PERCLOS** | Percentage of Eyelid Closure Over Pupil |
| **FOV** | Field of View | **PSNR** | Peak Signal-to-Noise Ratio (dB) |
| **FPS** | Frames Per Second | **REST** | Representational State Transfer |
| **GF** | Guided Filter / Fast Guided Filter | **ROI** | Region of Interest |
| **GFLOPs** | Giga Floating-Point Operations Per Sec | **SRS** | System Requirements Specification |
| **GPS** | Global Positioning System | **SSIM** | Structural Similarity Index Measure |